

Project X Version 4.0 - Technical Design Specification - Release 0

Vision

Project X preserves and evolves a teacher's professional knowledge throughout an entire career.

Architecture

Core, Packages, Plugins, Services, Data, UI Components.

Core Engines

Knowledge, Workspace, Storage, Search, Relationship, Asset, Academic Year, AI Context.

Object Model

AcademicYear, Curriculum, YearGroup, Unit, Lesson, TeachingAsset, Assessment, Evidence, Reflection, Observation, Resource.

ID Strategy

Universal immutable IDs such as PX-LSN-000001 and PX-AST-000001.

Storage

Storage abstraction supporting JSON today and database/cloud providers later.

Knowledge Graph

Relationships replace duplication.

Teaching Assets

All resources are managed assets linked to lessons.

Developer Standards

Reusable components, modular architecture, documentation-first, configuration-driven design.

Roadmap

Release 1: Knowledge Workspace; Release 2: Lesson Twin; Release 3: Asset Manager; Release 4: Academic Year Engine; Release 5: AI Context Engine.